Cheng Chen

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Education

Tongji University, Shanghai, China — Sept. 2023 – Present
B.Eng. in Electrical Engineering, College of Electronic and Information Engineering
Major Rank: **3/85**

Publications

- Yi Xu*, **Cheng Chen***, Wenzhuo Lei.
  If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization
  ACM MM 2026 (Accepted).
- **Cheng Chen***, Yi Xu*, Mufan Cao.
  OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment
  IJCAI 2026 (Accepted).
- Two papers under review at ***NeurIPS 2026***: Main Track and Evaluations & Datasets Track.

* equal contribution.

Research Experiences

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization

Accepted at **ACM MM 2026**, Co-first Author

- Addressed modality reliability asymmetry in open-vocabulary audio-visual event localization, where audio is semantically useful but often off-screen, delayed, or noisy, so symmetric distillation lets weak-audio supervision distort temporal boundaries.
- Proposed **OV-OrthKD**, a reliability-aware asymmetric distillation framework: a strong visual teacher supervises decision-aligned features, a weak audio teacher is restricted to an auxiliary orthogonal subspace, and a text prototype anchors open-vocabulary transfer.
- On OV-AVEBench, reached **0.816** segment AP and improved F1@0.5 by **+2.7** points overall and **+3.4** on unseen categories over the official fine-tuning baseline; ablations show weak-audio feature transfer helps only when kept off the decision path.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment

Accepted at **IJCAI 2026**, Co-first Author

- Addressed expertise asymmetry in multi-teacher distillation for diabetic retinopathy screening, where a weaker ViT teacher (**0.83 QWK**) harms students under standard logit fusion with a stronger CNN teacher (**0.88 QWK**).
- Proposed **OrthKD**, distilling only complementary feature-level context from the weak teacher while enforcing orthogonality between teacher projections to avoid redundant knowledge transfer.
- Evaluated on **132,049** retinal images; the **5.4M**-parameter MobileNetV3 student reached **0.885 QWK** on EyePACS and improved zero-shot Messidor-2 performance from **0.507** to **0.728 QWK** (**+22.1** percentage points).

Neural-Operator Reliability and Risk Triage

Research Project, First Author

- Addressed the deployment paradox of neural operators: single-pass surrogates are fast, but deciding when to trust a predicted field often requires multi-pass uncertainty estimators that erode the latency advantage.
- Proposed **AUD** (Allocation Uncertainty Decomposition), a single-pass risk score read from ambiguity in prediction-path allocation; in the wavelet instantiation **MWNO-AUD**, the same masks that modulate detail coefficients also yield local allocation entropy H_L.
- On EM scattering, H_L improved Spearman ρ_s from **0.61** to **0.67** and AUROC from **0.78** to **0.82** at **1.13**× batched cost; escalating only high-entropy cases raised ranking to $\rho_s = \mathbf{0.72}$ and AUROC=**0.85**.

Awards

- Provincial First Prize, China Robot and Artificial Intelligence Competition (2026), Team Captain
- National 2nd Prize, Global Campus AI Algorithm Elite Competition (2025), Team Captain
- National 3rd Prize, Embedded Chip & System Design Competition, FPGA track (2025), Team Captain

Skills

Programming: Python, PyTorch, Git, Linux, LaTeX, experiment tracking, LLM/agent prototyping.
Foundations: Machine Learning, Data Structures & Algorithms, Linear Algebra, Probability, Optimization, Information Theory, Signal Processing.
Passion: Deep enthusiasm for exploring the unknown and continuous learning across new areas.